

RoboVerse Qualifier Challenge

The Qualifier Challenge requires the teams to apply object detection and obstacle avoidance to execute missions without the use of Global Navigation Satellite System (GNSS).

Challenge Overview

The Challenge requires the team to navigate the quadcopter in a space port and search for fuel barrels. The teams are given 10 minutes. Within 10 minutes, the teams can make more than 1 attempt. The score will be based on the best attempt.

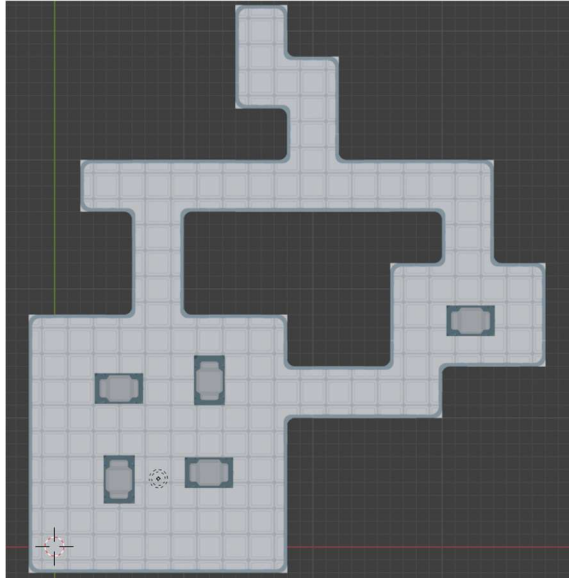


There are two types of fuel barrels: yellow and red. Red fuel barrels are not placed on ground level while yellow barrels are only placed on ground level.

University category teams must detect at least one red fuel barrel and at least one yellow fuel barrel in order for their scores to be eligible for Qualifier ranking. Pre-U Category is to detect at least one yellow or one red fuel barrel in order for their scores to be eligible for Qualifier ranking.

Challenge Map

The size of the space port is 40m (width) x 40m (breath) x 8m (high). Each grid is estimated to be 4m x 4m. The top down view of the space port is shown below. Locations of objects and grid layout may be different. The actual map will be released 1 day before the Qualifier.



Challenge Scoring

The judges award points for the following:

Criterion	Points
Detect a yellow fuel barrel (a barrel can only be detected once)	50
Detect red fuel barrel (a barrel can only be detected once)	100

The judges award bonus points based on the following timing:

Criterion	Points
Detects all yellow fuels within 5 minutes.	20 points per every 30 seconds in less than 5 minutes.

Detects all red fuels within 5 minutes.	20 points per every 30 seconds in less than 5 minutes.
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Teams who manually control quadcopter using a keyboard without/with mouse or using controller/joystick/gamepad/mouse will be disqualified.

Qualifier Process

Teams are to demonstrate the Qualifier challenge in person between 22 May 2026 and 23 May 2026. Teams are to book a session from one of the Qualifier sessions on 22 May 2026 and 23 May 2026. The coordinators will announce the timing of sessions and process to book a session. Teams are to arrive 15 minutes before the timing of their sessions.

Teams are given 15 minutes to set up and test their codes to run on the computers provided by the coordinators. Teams are to bring their codes on a removable drive.

Once ready, the team is to wait for the judge's instructions to start their run. When the run starts, the 10 minutes countdown starts. The judge will not stop the clock when the code crashes. The judge will only stop the clock when there are hardware issues with the computer.

When the 10 minutes is up, the judge will brief the best attempt scores and the team is to acknowledge the result.